# ZIP 98026, Edmonds and Esperance

*The winter danger here is not the storm. It is the grill, the oven, or the generator people use indoors on day two of a power cut. Then, in 2027, a second year of thin snowpack.*

| IMPACT REGION | EXCESS HEAT DEATHS | WINDOW | DOMINANT HAZARD |
|---|---|---|---|
| Snohomish County, WA (containing geography) | No figure published | Sep 2026 – Feb 2027 (6-month) | Outages and carbon monoxide; 2027 snow drought and smoke |

**28/28 recommendations checked.** 16 corrected, 2 unverified, 6 contraindicated. Exposure figure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review. It publishes no figure for this county. Winter heat is not this place's hazard, so the historical record carries the risk here instead of a heat number. El Niño is strong (ONI +2.16 for Jul–Sep 2026; NOAA CPC El Niño Advisory, 8 October), and the winter outlook favours above-normal temperatures for the Pacific Northwest.

EXECUTIVE SUMMARY — AT A GLANCE




47.79°N 122.35°W, United States of America · Natural Earth

- **The risk.** In the 2006 windstorm, 8 of about 15 Washington deaths were carbon monoxide poisonings from fuel burned indoors during the outage that followed.
- **For you.** By 15 November, fit a carbon monoxide alarm on every sleeping floor ($20–40 each), and never run a grill, stove, or generator inside.*
- **The money.** $20k floor, $53k target, $63k ceiling. Most is buying CO alarms that South County Fire will install but has none of.
- **Do not.** Run a generator in a garage, shed, or porch, even with the door open. Keep it 20 feet from any opening.*


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* Suggested formulation: check with a health worker before sharing.

FOR YOU, AS A HOUSEHOLD 8 STEPS

For a household on the Esperance plateau, El Niño probably makes this winter milder, not more dangerous. Windstorms and cold snaps become less likely, but neither stops: the 2006 storm and the January 2024 freeze both came in El Niño winters. What hurts families here is a long power cut, five to seven days in the November 2024 storm, and what people do to stay warm and fed while it lasts. Next summer brings the second risk: low water and wildfire smoke. The ranked actions below are written for the county, the fire district, and their partners. These eight are the ones you can take yourself, grouped by what they protect.

### 1 · Carbon monoxide: fit a battery alarm outside each sleeping area and on every level by 15 November, and make "nothing that burns fuel runs indoors" a household rule

CONFIRMED FROM ACTIONS 1 AND 2

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 15 Nov 2026 | $20–40 per alarm (UL 2034 listed) | Everyone asleep in the house |

This is the step with the strongest local evidence. In King County after the 2006 storm there were 259 poisonings and 8 deaths, all in homes where a generator or charcoal was burning in an enclosed space. In a Wisconsin study, people without a working alarm had 3.2 times the odds of being poisoned. The rule covers generators, grills, camp stoves, propane heaters, and using the oven or gas range for heat. Keep a generator outdoors, at least 20 feet from doors, windows, and vents (CDC). Washington has required CO alarms in homes since 2013. If you rent, your landlord must provide one and you must keep its batteries fresh. South County Fire will install an alarm you have bought, free.*

### 2 · Power: plan to be without electricity for a week, and sign up now for the two alert systems that will tell you what is happening

PARTIALLY CONFIRMED FROM ACTION 4

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 15 Nov 2026 | $0 for alerts; about $30–150 for lights and a power bank (estimate) | Medical devices, phones, information |

* Suggested formulation: check with a health worker before sharing.

Register your address with **SnoCoAlerts**, the county's alert system (snohomishcountywa.gov/620). It sends by address, which matters because Esperance is not part of the City of Edmonds. Turn on outage texts in your MySnoPUD account (outages: 425-783-1001). In November 2024, 135,000 PUD customers lost power and it took about six days to restore 99% of them. Keep flashlights and headlamps, not candles, and a charged power bank for each phone. If anyone uses an oxygen concentrator, CPAP, or other powered device, arrange your own backup battery and a place to go with power. The PUD has no published medical priority registry, so do not count on being restored first.*

### 3 · Staying warm: choose one room to heat and close off the rest, and save the county's cold-weather page now, while you still have power

CONFIRMED FROM ACTION 4

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| Before the first storm | $0–50 (blankets, sleeping bags) | Older and very young family members |

Choose a small room, ideally with few outside walls. Hang blankets over its doorways and gather everyone there in layers and sleeping bags. If it is still too cold, leave rather than burn something indoors. Overnight shelters open when the forecast is 34°F or below: Maple Park Lutheran in Lynnwood, run by Volunteers of America, intake 7–10pm. Community Transit rides are free on those nights. The Edmonds Library Plaza Room is a daytime warming site. Check what is open at snohd.org/cold.*

### 4 · Food: keep five days of food that needs no cooking and no fridge, and put thermometers in the fridge and freezer by 1 December

CONFIRMED FROM ACTION 7

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 1 Dec 2026 | $40–60 for thermometers and a cooler; about $50–75 a person for food (estimates) | Your food budget, and stomachs |

Outages, not prices, are where the food chain breaks here: in the home fridge. USDA's rule is that a closed fridge keeps food safe for about 4 hours, a full freezer for 48, and a half-full one for 24. Throw out perishables that have been above 40°F for more than 2 hours.

* Suggested formulation: check with a health worker before sharing.

Food prices are already rising; Seattle-area food-at-home prices were up 4.5% on a year in August. That is a reason to shop steadily, not to stockpile.*

### 5 · Water: store a gallon per person per day for two weeks, and find out which utility supplies your tap

PARTIALLY CONFIRMED | FROM ACTION 8

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 1 Dec 2026; then from 1 Apr 2027 | About $20–40 for containers (estimate) | Drinking water and pets |

Washington's "2 Weeks Ready" guidance is 1 gallon per person, and per pet, per day: 42 gallons for three people. Your water bill will show the utility. In and around Esperance it is most likely Olympic View Water & Sewer District, which blends its own Deer Creek supply with Seattle Public Utilities water, not Everett's. So a 2027 drought request would follow Seattle's plan. Washington is already in its fourth statewide drought since 2015, after the snowpack failed in 2026. An El Niño winter usually means a low snowpack again by 1 April. From April, watch your utility's notices and be ready to cut use by about 10%.*

### 6 · Communications: a battery radio, an out-of-area contact, and a list of the 15–20 homes around you

PARTIALLY CONFIRMED | ROLE-SPECIFIC

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 15 Nov 2026 | About $25–60 for a NOAA weather radio (estimate) | Contact when phones and internet fail |

In the November 2024 storm, some agencies could not post updates because their own internet was down. A battery or hand-crank NOAA weather radio still works. Agree on one relative out of state whom everyone calls or texts, because long-distance texts often get through when local calls fail. Then use the state's Map Your Neighborhood method: meet the 15–20 homes nearest you, note who lives alone, who uses powered medical equipment, and who has no car, and agree who checks on whom after a storm. The state stopped printing the workbook in 2021; a list on paper is enough.

### 7 · Health: flu, RSV, and COVID shots by 15 November, a week of prescriptions on hand, and a smoke filter built before next summer

* Suggested formulation: check with a health worker before sharing.

PARTIALLY CONFIRMED · FROM ACTIONS 5 AND 8

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 15 Nov 2026; filter by 1 Jun 2027 | Usually $0 for insured shots; $60–100 for a filter | Lungs and hearts, especially over 65 |

Respiratory illness, not El Niño, is the most reliable winter killer of older adults here. Get a flu shot for everyone over 6 months, the high-dose or adjuvanted kind for anyone 65 or over. Get *one lifetime* RSV dose for anyone 75 or over, or 50 to 74 with a heart, lung, or other qualifying condition, if they have not had it. In its first season it cut severe RSV disease by about 94% in adults 60 and over. Washington recommends the 2026–27 COVID vaccine for all adults. On Medicare, the RSV shot is $0 only through Part D. For summer 2027 smoke, build a box-fan filter from four MERV-13 filters, or buy a HEPA cleaner sized for the room by its clean-air delivery rate. Either should be CARB-certified or have no ionizer.*

### 8 · Money: check the PUD discount this week, apply for energy help between 1 and 10 November if you qualify, and read your insurance policy for what it leaves out

PARTIALLY CONFIRMED · FROM ACTION 6

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 1–10 Nov 2026; flood cover before 11 Dec | $0 to apply | Heat, bills, and the house |

The Snohomish PUD takes 25% or 50% off future bills for income-qualified households: up to $43,280 a year for two people, $66,000 for four (snopud.com/discount, 425-783-1000). The county's energy-assistance office takes applications from everyone eligible only from 1 to 10 November; grants run $250–1,250 ((425) 388-3880, weekdays 9–11:30). From 15 November to 15 March, a PUD or PSE gas customer who calls within five business days of a shutoff notice can stay connected on a payment plan capped at 7% of income. Standard homeowner policies in Washington exclude landslide and earth movement, and most exclude flood. If you want flood cover, buy it well before 11 December: the federal flood programme's authority runs out with the current stopgap budget, and a new policy takes 30 days to start. On the plateau, flood risk is low; earth movement and sewer backup matter more.

## What to ask of the people in charge

* Suggested formulation: check with a health worker before sharing.

- Ask **Snohomish County Emergency Management** whether Esperance addresses are on the warming-site notices and welfare-check lists, not only on SnoCoAlerts (action 3).
- Ask **South County Fire** for a free home safety visit, and whether CO alarms for neighbours who cannot afford them are coming this winter (action 1).
- Ask **Olympic View Water & Sewer** (425-774-7769) which supply your tap is on, and what reading would trigger a 2027 request to cut use (action 8).

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

## 1. By 1 December, check, fit, and replace carbon monoxide alarms in the homes least likely to have a working one: the mobile-home parks, older owner-occupied houses, and Highway 99 apartments

PARTIALLY CONFIRMED | EMERGENCY MANAGER · PUBLIC HEALTH · FRONTLINE · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| South County Fire, funded by a local grant | 300 by 1 Dec; up to 1,000 by 28 Feb 2027 | About $9,000 for 300; $30,000 for 1,000 (alarms at ~$30, estimate) | CO poisoning during multi-day outages |

South County Fire runs free home safety visits, and its own page says it will install CO alarms but has none to give out. The gap is that simple. The law is not the gap: RCW 19.27.530 has required CO alarms in residences since 2013, including rentals and manufactured homes. It exempts owner-occupied houses lived in since before July 2009 and never sold, and it does not keep batteries alive. So the visit should test and replace, not assume a home has none. Where a rental lacks an alarm, the visit should point the landlord to the law rather than spend grant money. Target Trailer Lane (23515 Hwy 99), Woodruffs (7806 218th St SW), and Ridge Acres (23121 Edmonds Way, ZIP 98020), and the corridor apartments. Evidence: Wisconsin 2014–16, odds of poisoning 3.2 times higher without an alarm that sounded (95% CI 1.5–6.9). In New Mexico, 1980–95, alarms might have prevented about half of 136 unintentional non-fire CO deaths (Yoon, JAMA 1998; modelled).

* Suggested formulation: check with a health worker before sharing.

Transferability: the deaths this targets happened in this region (King County, 2006), in the same kind of homes. The district's largest recorded single install was 62 alarms (2017), so start with 300 and scale only if it keeps pace.

## 2. By 15 November, pre-write carbon monoxide warnings in the corridor's languages, queue them in SnoCoAlerts to send early in any large outage, and hand printed cards out through the places people trust

PARTIALLY CONFIRMED | EMERGENCY MANAGER · FRONTLINE · SKEPTIC · PUBLIC HEALTH · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Snohomish County DEM, with the Snohomish Health District | 15 Nov 2026 | Under $10,000 for translation (lens estimate, not checked) plus staff time | The 2006 pattern: poisonings in households the warning never reached |

Write the messages now in Spanish, Korean, Vietnamese, Somali, Amharic, Russian, Ukrainian, and Chinese. Somali, Spanish, and Vietnamese were the commonest languages among the 2006 cases. Queue them to go out within the first hours of any outage of more than about 10,000 PUD customers; that trigger is a proposal, not an existing protocol. Then deliver them by hand through the Korean Community Service Center (22727 Hwy 99), St. Michael Ethiopian Orthodox Church (23010 84th Ave W), LETI in Lynnwood, the Highway 99 grocers, and the Edmonds Food Bank line. Evidence: in King County in 2006, 58% of poisoned households were immigrant households and 82% of those had used charcoal. All 8 deaths were in four immigrant households (Gulati, AJPH 2009). The same paper notes that evidence on whether education campaigns work is limited.

* Suggested formulation: check with a health worker before sharing.

Transferability: the local population is similar, and Highway 99 is home to many of the same communities. Whether SnoCoAlerts supports all eight languages was not checked.

### 3. By 1 November, confirm that unincorporated Esperance is inside every list the response uses: warming-site notices, welfare checks, and the City of Edmonds' own messages

CONFIRMED | EMERGENCY MANAGER · FRONTLINE | ADAPTATION INVENTORY: NOT CONSULTED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Snohomish County DEM, with the City of Edmonds and South County Fire | 1 Nov 2026 | None | An enclave falling between city and county |

Esperance is a county enclave surrounded by Edmonds. Esperance voters have rejected annexation five times, most recently in 2005. Talks reopened after December 2025, with a first council fiscal review in July 2026. SnoCoAlerts targets by address, so it should reach Esperance. Whether the city's own outreach lists include it is not confirmed. South County Fire covers both the city and Esperance.

Transferability: this is a local administrative check, not borrowed evidence.

### 4. By 1 December, open warming sites when an area has been without power for 24 hours, not only when the forecast hits 34°F, and route outage check-ins through the Sheriff's existing call programme

PARTIALLY CONFIRMED | EMERGENCY MANAGER · PUBLIC HEALTH · FRONTLINE · HISTORIAN | ADAPTATION INVENTORY: NOT CONSULTED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Snohomish County DEM, with Sno-Isle Libraries and Volunteers of America | 1 Dec 2026 | Existing contracts; generator rental not costed | Cold homes on day two, where the grill comes indoors |

The system today: shelters open at 34°F or below, with status at snohd.org/cold. Maple Park Lutheran (17620 60th Ave W, Lynnwood) is run by Volunteers of America, intake 7–10pm, with free Community Transit rides. The Edmonds Library Plaza Room is a daytime warming site. Most windstorm outages here come in mild weather, so a temperature trigger alone misses them. In Texas in 2021, most of 246

* Suggested formulation: check with a health worker before sharing.

storm deaths were hypothermia in homes without power, and 8% were carbon monoxide. For check-ins, the Sheriff's Office already runs "You Are Not Alone", scheduled calls and welfare checks for seniors who live alone. Extend it to outage days, and use HHS emPOWER's ZIP-level counts of Medicare users on powered devices to size it. Do not build a separate register: one cannot be built by 1 December.

Transferability: the evidence that living alone raises risk comes from Chicago heat deaths (Semenza, 1996, odds ratio 2.3), not cold. That it applies here is inference.

## 5. By 15 November, run flu, RSV, and COVID vaccination at the senior centre and pharmacies for adults 65 and over, putting the one-time RSV dose first for those 75 and over

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Snohomish Health District, with pharmacies and the Edmonds Waterfront Center | 15 Nov 2026 | Vaccine covered by insurance; clinic staff time | The winter's most dependable deaths among older adults |

About 22–24% of Edmonds residents are 65 or over; QuickFacts could not be re-read to confirm. RSV vaccine is one lifetime dose for all adults 75 and over and for adults 50–74 at increased risk, the age floor having dropped from 60 in 2025. Only 40.9% of adults 75 and over nationally have had it. In the AReSVi-006 trial of about 25,000 adults 60 and over (Papi, NEJM 2023), it was 82.6% effective against RSV lower-respiratory disease and 94.1% against severe disease in the first season. Protection falls in later seasons, and FDA added a Guillain-Barré warning in January 2025. Washington's 2026–27 schedule follows the West Coast Health Alliance and recommends COVID vaccine for all adults 19 and over. Insurers have pledged $0 coverage only through the end of 2026.*

* Suggested formulation: check with a health worker before sharing.

Transferability: good, since the trial population was older adults in high-income countries. This action is not El Niño-specific. It is here because El Niño probably *lowers* this winter's storm and cold risk while respiratory illness comes every winter.

## 6. During 1–10 November, get every eligible household's energy-assistance application in, enrol them in the PUD discount, and tell them about the winter shutoff rule

PARTIALLY CONFIRMED | ECONOMIST · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CONSULTED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Snohomish County Office of Weatherization & Energy Assistance, with Snohomish PUD and PSE | 10 Nov 2026 (open window); 15 Nov (shutoff rule starts) | Staff time; grants $250–1,250 per household from LIHEAP and SHEAP | Households heating with whatever they can burn because the bill is unpaid |

Three instruments already exist, and each needs the household to act. LIHEAP: the 1–10 November window is the only one open to everyone eligible; after it, only crisis cases and PSE HELP. The PUD discount: 25% or 50% off future bills. DSHS clients and free-lunch families get streamlined approval, but nobody is enrolled automatically. The winter shutoff protection: under RCW 54.16.285 for the PUD and WAC 480-90-143 for PSE gas, no disconnection from 15 November to 15 March for a customer who calls within five business days of the notice. Payments are capped at 7% of income plus a twelfth of arrears, with the balance due by 15 October. Federal LIHEAP runs on a stopgap to 11 December 2026, and the White House has again proposed ending it for FY2027.

* Suggested formulation: check with a health worker before sharing.

Payer and trigger: HHS through WA Commerce and the county (on application); PUD ratepayers (on approval); none for the shutoff rule, which is forbearance released by one phone call. It goes unused because few people know the call has to be made.

## 7. By 1 December, give the Edmonds Food Bank a generator hookup and a stock of no-cook outage boxes, and by 31 December screen Basic Food recipients for work-rule exemptions

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CONSULTED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| Edmonds Food Bank, with DSHS Community Services Offices | 1 Dec 2026; screening by 31 Dec | Unfunded: generator and boxes not costed to a source | Lost cold stock in outage week; an April 2027 benefit cliff |

The food bank serves 1,200–1,500 households a week, a record in its 40 years, from 828 Caspers St. It is raising $12M for a new building and has no recorded backup power. The PUD restores "critical businesses" first but publishes no list to join, so the hookup is the realistic step; Monroe Food Bank's rooftop solar is a local precedent. On SNAP: since 4 July 2025 the work-requirement age runs to 64, and Washington has had no waived areas since 1 February 2026. Losses are already landing, and a new 36-month period starts 1 January 2027. About 72,000–78,000 county residents receive Basic Food. Do not stockpile rice and oil ahead of a lagged El Niño price peak. The rise is already here (FAO, September 2026: vegetable oils +18.3%, cereals +17.2% on a year), and most US rice is US-grown. Cash for wholesale buying beats a speculative stock.

* Suggested formulation: check with a health worker before sharing.

Trigger to watch: BLS Seattle-Tacoma-Bellevue CPI, food at home, published every two months (+4.5% on a year in August; next release 10 November).

## 8. By 1 February, Olympic View and the south-county water utilities agree spring drought messages tied to the 1 April snowpack reading, and smoke filters are in place by 1 June 2027

PARTIALLY CONFIRMED | HISTORIAN · EMERGENCY MANAGER · PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CONSULTED

| OWNER | BY WHEN | COST | PREVENTS |
|---|---|---|---|
| **Olympic View Water & Sewer District, with Seattle Public Utilities and the Snohomish Health District** | **1 Feb 2027; filters by 1 Jun 2027** | **None for messaging; filters ~$85 each, recipient not named** | **A late summer scramble, as in 2015** |

Washington declared a statewide drought on 8 April 2026, the fourth since 2015, after a wet winter fell as rain and left about half the usual snowpack. Seattle, Everett, and Tacoma said they expected no impact on their customers. A strong El Niño now follows, and El Niño winters here usually leave the 1 April snowpack below normal. In 2015, after a weaker El Niño winter, Everett activated its drought plan for the first time and asked for 10% cuts. Olympic View blends its own supply with Seattle water, so Seattle's plan sets its stage. For smoke: a box fan with four MERV-13 filters delivers a clean-air rate of about 600–850 cubic feet a minute (Dal Porto, 2022). HEPA filtration in a woodsmoke town cut indoor fine particles 60% (Allen, 2011, healthy adults, short-term markers only).

Transferability: the snowpack link is a risk signal, not a forecast. Smoke here also depends on fires in BC and eastern Washington. A very strong El Niño can also break the pattern and bring a wet winter.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

## RACI CHART

**R** = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

| Action | South County Fire | Sno. Co. DEM | Health District | Hwy 99 hubs | Shelter operators | OWEA, PUD & PSE | Edmonds Food Bank | Olympic View W&S | City of Edmonds | Local funders |
|---|---|---|---|---|---|---|---|---|---|---|
| 1. Check and fit CO alarms | **A/R** | **I** | **C** | **C** | | | **I** | | **I** | **R** |
| 2. Multilingual CO warnings | **C** | **A** | **R** | **R** | | **C** | **R** | | **I** | **C** |
| 3. Esperance on every list | **C** | **A/R** | | | **I** | | | | **R** | |
| 4. Outage trigger for warming sites | **C** | **A** | **C** | **I** | **R** | **C** | | | **R** | |
| 5. Flu, RSV, and COVID shots | | **I** | **A/R** | **C** | | | | | **I** | |
| 6. Energy assistance and shutoff rule | | **I** | | **R** | | **A/R** | **C** | | **I** | |
| 7. Food bank power and SNAP screening | | **C** | | **C** | | **C** | **A/R** | | **I** | **R** |
| 8. 2027 drought messages and smoke filters | | **C** | **R** | **I** | | | | **A/R** | **I** | **C** |

What the chart shows: Snohomish County Emergency Management is accountable for three of the eight rows, more than any other owner. South County Fire carries the single highest-value row. Actions 3 and 4 need a decision, not money. Actions 1 and 7 wait on money: action 1 on a grant, action 7 on funding nobody has costed yet.

WHERE MONEY DOES THE MOST

## FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

**$20,000: 500 alarms and the warnings**

TARGET

**$53,000: sum of the verified placements**

CEILING

**$63,000: limited by installers, not money**

LARGEST UNCERTAINTY

**How many alarms South County Fire can fit per week**

Floor = 500 alarms × $30 ($15,000) + $5,000 for warnings. Target = $30,000 (alarms, staged) + $5,000 (warnings) + $10,000 (check-in pilot) + $8,000 (vouchers, if not already funded) = $53,000. Ceiling = target + $10,000 for the PUD Community Energy Fund = $63,000. The target fell from $68,000 in verification (−22%): the warnings grant was halved, the check-in coordinator cut to a pilot, and smoke filters dropped until someone is named to receive them. Every unit price is an estimate; the $30 alarm assumes a bulk price, since retail runs $22–40. The fastest local funder, the Hazel Miller Foundation, reviews applications quarterly, so new money would arrive around February. That is late for the floor, which is why the floor should come from a donor or an existing discretionary budget.

* Suggested formulation: check with a health worker before sharing.

| RECIPIENT | USE | AMOUNT | MECHANISM | BASIS | VERDICT |
|---|---|---|---|---|---|
| South County Fire | Action 1: CO alarms fitted at home safety visits, 300 first, up to 1,000 | $30,000 | Restricted grant, staged | ~$30 per alarm (estimate); alarms "may have prevented approximately half" of non-fire CO deaths (Yoon, JAMA 1998, New Mexico) | Partially confirmed |
| Edmonds Food Bank | Action 2: printing and hand distribution of the multilingual cards | $5,000 | Pass-through grant | Gulati, AJPH 2009, on who was poisoned; no measured effect of a campaign exists | Partially confirmed |
| Edmonds Waterfront Center (senior centre) | Action 4: outage-day check-in pilot, coordinated with the Sheriff's "You Are Not Alone" programme | $10,000 | Direct grant, pilot | Staff cost only; no published deaths-averted figure | Unverified |
| Volunteers of America Western Washington | Action 4: hotel vouchers for medically fragile shelter guests | $8,000 | Top-up, only if its county award has no voucher line | County RFP: vouchers "typically range between $5,000 to $8,000 annually" | Partially confirmed |
| St. Vincent de Paul, PUD Community Energy Fund (formerly Project PRIDE) | Action 6: bill help for households that miss LIHEAP (ceiling only) | $10,000 | Top-up | 100% goes to assistance; overlaps with the PUD discount and LIHEAP | Partially confirmed |
| Not yet named | Action 8: 200 box-fan filters for summer 2027 smoke | $17,000 (not counted) | Pre-positioned stock | ~$85 per box; no recipient or existing programme found | Unverified |

The funder's one thing: the visible need, shelter nights, is already paid for. The county funds the cold-weather shelter network at about $931,000 for September 2026 to September 2027, and the Lynnwood site opened only 25 nights last winter. The deaths in this place's record happened inside heated homes after windstorms. The cheapest life saved here is a $30 alarm someone else installs.

* Suggested formulation: check with a health worker before sharing.

# Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

## Do not run a generator in a garage, shed, carport, or porch, or heat with a grill, camp heater, oven, or gas range, even with a door open *

CONTRAINDICATED

This is what killed people here. In King County after the December 2006 storm, all 8 CO deaths happened in homes where a generator or charcoal was burning in an enclosed space. Four of the dead were one family in Burien, with the generator in the garage. A Lynnwood fourplex burned from a generator running inside a shed. In the 2021 Texas freeze, more than 1,400 people were treated for CO poisoning.

**Instead:** run a generator outdoors, at least 20 feet from doors, windows, and vents (CDC), with a working CO alarm inside. If the house is too cold, go to a warming site.

## Do not rely on English-only, web-only warnings sent once the outage has begun

CONTRAINDICATED

In 2006 the poisonings were concentrated in households with Somali, Spanish, or Vietnamese as their first language, and most patients arrived within 48 hours of the storm. A warning written during the outage arrives after the grill has come inside. Web-only messages fail when the power and the internet are down; in November 2024 some agencies could not post at all.

* Suggested formulation: check with a health worker before sharing.

**Instead:** write the messages now, in the corridor's languages, and put them in people's hands through the churches, grocers, and community centres on Highway 99 (action 2).

### Do not skip winter preparation because "El Niño means a mild winter"

CONTRAINDICATED

El Niño lowers the odds of windstorms and cold snaps here; it does not remove them. The Hanukkah Eve windstorm of December 2006 came in an El Niño winter. So did the January 2024 arctic outbreak, in which at least five people died of hypothermia in King County and shelters neared capacity.

**Instead:** prepare for a week without power in a cold spell, as in the For you steps, whatever the outlook says.

### Do not buy an ozone generator or an ionizing "purifier" for wildfire smoke

CONTRAINDICATED

These add ozone, a lung irritant, to the air they are meant to clean. CARB certification shows a unit is low in ozone; it says nothing about how well it cleans.

**Instead:** a HEPA cleaner sized by its clean-air delivery rate, or a box fan with four MERV-13 filters, bought in spring before smoke arrives and stores sell out.*

### Do not spend new money on more shelter nights as the El Niño response

CONTRAINDICATED

The county already funds the cold-weather shelter network for the whole season, and a warm-leaning El Niño winter means fewer nights below 34°F to open them. New money there duplicates committed money, and it does nothing for the people most at risk: those in their own homes, with no power, burning fuel to keep warm.

* Suggested formulation: check with a health worker before sharing.

**Instead:** fund alarms (action 1), and change the trigger so warming sites open for long outages too (action 4).

### Do not stockpile rice and cooking oil ahead of a "lagged" El Niño price peak

CONTRAINDICATED

The price rise is already here: global vegetable oil prices were up 18.3% on a year in September 2026, and US retail rice is mostly US-grown. Buying months of stock now ties up a household's or a food bank's cash at today's high prices, for a peak that may not come.

**Instead:** keep five days of no-cook food for outages (For you, step 4). Give food banks cash, not tins, so they can buy at wholesale.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| Jan 1997 (neutral) | Woodway bluff landslide, about 100,000 m³, derails five BNSF freight cars two hours after an Amtrak train with 650 aboard had passed | No deaths | Later: catchment walls and groundwater monitors on the Mukilteo reach cut passenger-rail cancellations from 200+ (2012) to about 53 a year |
| Dec 2006 (El Niño, weak) | Hanukkah Eve windstorm; more than 1.5 million customers out in the region, some for up to 11 days | About 14–15 Washington deaths, 8 of them carbon monoxide (King County); 259 CO poisonings in King County | King County's later review found translation alone was not enough |
| 2015 (after the weak 2014–15 El Niño) | Snow drought: record-low snowpack, then a hot summer | Farm losses of $633–773 million statewide | Everett activated its 2001 drought plan for the first time; October rains refilled Spada Lake |
| Jun 2021 (La Niña / neutral) | Heat dome | 12 heat deaths in Snohomish County confirmed by the Medical Examiner; 15 for the whole summer (state, preliminary) | Cooling centres added; county ARPA money put $1.5 million into heating and cooling upgrades at libraries and senior centres |

* Suggested formulation: check with a health worker before sharing.

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| Dec 2022 (La Niña) | King tide floods the Edmonds waterfront, about 18 inches, while the road stayed open | No deaths reported | Nothing recorded; no waterfront adaptation plan found |
| Jan 2024 (El Niño, strong) | Arctic outbreak | At least 5 hypothermia deaths in King County | Not found |
| Nov 2024 (neutral) | Bomb cyclone; 135,000 Snohomish PUD customers out, about six days to restore 99% | A woman killed by a falling tree at a Lynnwood encampment | Not found |
| Apr 2026 (La Niña winter) | Statewide drought declared after a rain-not-snow winter left about half the usual snowpack | Not applicable | Seattle, Everett, and Tacoma expected no customer impact |

The recurrence nobody has noticed: back-to-back snow droughts. Most of the famous storms here were not El Niño events at all. What El Niño reliably brings is a thin snowpack, and 2026 already had one. The 2023 drought ran straight into 2024. A strong El Niño now follows a failed snow year, and the region's buffer is a few reservoirs refilled by autumn rain.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Prepare for this winter's storm** versus **El Niño lowers this winter's acute risk; the real El Niño hazard is 2027's water and smoke.** The emergency manager and frontline lenses hold the first view; the public health scientist and historian the second. Both lead to the same first actions, because carbon monoxide and respiratory illness recur every winter. What would settle it is the 1 April 2027 snowpack.
- **A strong El Niño means a dry, low-snow winter** versus **very strong El Niños "don't follow the typical rule of thumb" and can run wet.** NOAA's outlook favours warm here, with its wet tilt stopping at the Oregon coast. The drought actions should key off measured snowpack, not the forecast.
- **Buy staples now ahead of a lagged price peak** (food supply chain lens) versus **the rise is already here, so cash beats stockpiles** (verification). The verification view is adopted above.

* Suggested formulation: check with a health worker before sharing.

- **Cold shelter matters** (the January 2024 hypothermia deaths) versus **shelter nights are already funded** (philanthropist). Both hold. The gap is the outage trigger, not the beds.

## WHAT NOBODY COVERED — BLIND SPOT

Two gaps. First, people without homes along Highway 99: the panel noted the Lynnwood encampment death and the shelter network, but no lens mapped where unhoused people actually stay in 98026, or who reaches them before a windstorm. Second, earthquake. It is not an El Niño hazard, but it is this household's largest uninsured risk: only about 14% of homes west of the Cascades carry earthquake cover. The two-week kit in the For you steps serves both, which is a reason to build it fully rather than for one winter.

## HOW TO READ THIS — METHOD

**The panel is author-built.** Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

## HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The request gave a street address in ZIP 98026. It was widened to ZIP 98026, the smallest unit with data that still means something. The Census geocoder and Snohomish County's GIS place the address in unincorporated Esperance (Snohomish County), not the City of Edmonds, so the briefing covers both. Climate Impact Lab impact region: not resolved in the tool. The briefing uses Snohomish County as the containing geography. Mortality figures exist only at that scale; sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February

* Suggested formulation: check with a health worker before sharing.

2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.

- **Expert panel.** Eight simulated expert lenses each researched this place independently and in parallel, from web sources, to a fixed brief: emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). The six covered carbon monoxide; outages and cold; health; money; food and water; and the funding envelope. Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Could not be consulted. It is a client-side app that loaded showing "0 adaptations", and the download was not reached in this run. No action here should be read as lacking a precedent; none was checked.
- **Reader.** The requester's address is a home, so the reader was taken to be a household. The For you section is fuller than usual, with eight steps covering power, warmth, food, water, communications, health, and money. They were taken from the verified actions and from what the panel found about households here. Any step not drawn from a ranked action was verified on the same three tests. The rest of the report is written for the county, the fire district, and their partners.
- **Health wording.** Passages where a wording error could hurt someone (carbon monoxide, generators and grills, keeping warm in an outage, food and water in an outage, vaccines, and wildfire smoke) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing

* Suggested formulation: check with a health worker before sharing.

them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Census QuickFacts, the Snohomish PUD site, and Olympic View's site could not be read directly, so the age share, the absence of a PUD medical registry, and Olympic View's current supply mix are unconfirmed. Household cost ranges marked "estimate" are retail estimates, not quotes.

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References · verification status in brackets.

Carbon monoxide · Gulati et al., AJPH 2009, PMC2724456 [confirmed; "six-language warnings on Dec 20" not found in it] · Wisconsin detector study, 2014–16 [confirmed] · Yoon et al., JAMA 1998 [corrected: "approximately half"] · RCW 19.27.530 [confirmed] · South County Fire home safety visits [confirmed] · CDC generator safety [confirmed] · HistoryLink, 2006 storm · UW Climate Office [confirmed]

Outages and cold · SnoCoAlerts · Snohomish PUD outage centre · Herald, Nov 2024 restoration [corrected: ~6 days] · Lynnwood encampment death · cold-weather shelters · Sheriff "You Are Not Alone" · Map Your Neighborhood · 2 Weeks Ready · HHS emPOWER · Texas DSHS 2021 storm mortality · KUOW, Jan 2024 [lens source, not re-checked] · Semenza, NEJM 1996 [confirmed, heat]

Jurisdiction · Census geocoder · Snohomish County SAS Cities · My Edmonds News, Esperance · South County Fire [confirmed]

Climate · NOAA ONI · CPC ENSO discussion · CPC seasonal outlook · State Climatologist on El Niño · OPB, June 2026 [confirmed]

Health · Papi et al., NEJM 2023;388:595 [confirmed, first season] · Immunize.org RSV recommendations · WA DOH 2026–27 schedule · CIDRAP, AHIP coverage · Medicare Interactive · Allen et al. 2011, NCT01256957 [confirmed] · Dal Porto 2022, box-fan filters · CARB air cleaners · Snohomish County ME, 2021 heat · WA DOH, 2021 heat [both confirmed, different scopes]

Money · PUD discount · RCW 54.16.285 · WAC 480-90-143 · County energy assistance · APPA, LIHEAP FY27 · WA OIC, landslide insurance · NAHB, NFIP extension [confirmed] · County cold-weather shelter FAQ [confirmed]

Food and water · USDA ERS Food Price Outlook · FAO Food Price Index · BLS Seattle CPI · DSHS ABAWD · WSR 26-05-015 · Edmonds Food Bank · USDA FSIS outages [page blocked; standard guidance] · Ecology, April 2026 drought · Everett 2015 drought · Olympic View · SPU service area [supply mix unconfirmed] · USGS, 1997 landslides · Herald, rail landslides · NOAA, Dec 2022 water levels

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory



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